

Week 3 Introduction

Mathematics for Multidimensional Data



Video: 3.1 Multidimensional
Data Points and Features

2 min



Reading: Multidimensional
Data Points and Features
Recap

10 min



Practice Quiz:
Multidimensional Data
Points and Features –
Review Information

1 question



Video: 3.2 Multidimensional
Mean

2 min



Reading: Multidimensional
Mean Recap

10 min



Practice Quiz:
Multidimensional Mean –
Review Information

1 question



Video: 3.3 Dispersion:
Multidimensional Variables

3 min



Reading: Multidimensional
Variables Recap

10 min



Practice Quiz: Dispersion:
Multidimensional Variables
– Review Information

3 questions



Video: 3.4 Distance Metrics

5 min



Reading: Distance Metrics
Recap

10 min



Practice Quiz: Distance
Metrics – Review
Information

4 questions



Multidimensional Mean Recap

Don't mix up the letters (once more)!

Remember, when we talk about the **population** (all data points), the mean is denoted by the Greek letter μ , and we divide by the total number of elements N , and the standard deviation is denoted by the Greek letter σ . However, when we work with a **sample**, we use $\bar{x}$ as a name for the mean, we divide by n , and our standard deviation will be called s (and in this case, don't forget, we divide by $(n - 1)$ in order to calculate the estimate).

In vector form, when we pack everything into arrays for more efficient calculation, we obtain the following:

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{pmatrix} \text{ (the **population** mean vector), and}$$

$$\bar{\mathbf{X}} = \begin{pmatrix} \bar{X}_1 \\ \bar{X}_2 \\ \bar{X}_3 \end{pmatrix} \text{ (the **sample** mean vector).}$$

Attention!

In the previous video, at 0'56, since we estimate the population **mean** through a sample, and not the **variance** or **standard deviation**, we divide by n ($(n - 1)$ is necessary only for the variance or standard deviation).

As earlier in the course, the notion of **bias** refers to the fact that if you try and estimate information about the **population** by doing calculation only on a **sample**, your calculation may not lead to an approximation of the desired value. Hence this distinction that you need to remember:

- for the **mean**, as seen in the previous video, the estimate is **unbiased**, which means dividing by n will give you the right result;
- but, as mentioned already, if you are trying to estimate the **variance** or **standard deviation**, you have to use Bessel's correction, that is, you have to divide by $(n - 1)$ instead of only by n , which you could do if you had all the elements of your population available.